



A Risk-Taking Comparative Measure: Manual BART Testing



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BACKGROUND

- Risk-taking is a conceptual pattern of behavior that involves taking chances in situations with uncertain outcomes, especially in situations that involve possible negative consequences. (Leigh, 1999)
- Risk-taking behavior has important consequences for individuals (e.g., gambling and drug abuse) as well as for society at large (e.g., drunk driving deaths)
- The original BART task (Lejuez et al., 2002) is a computerized program in which people are asked to blow up a balloon as much as possible without popping it. Participants are incentivized with money as every time they pump the balloon, they earn more of it. If the balloon does pop, all money is lost for that balloon.

RESEARCH QUESTION

Is a manual version of the BART an accurate measure of risk-taking behavior?

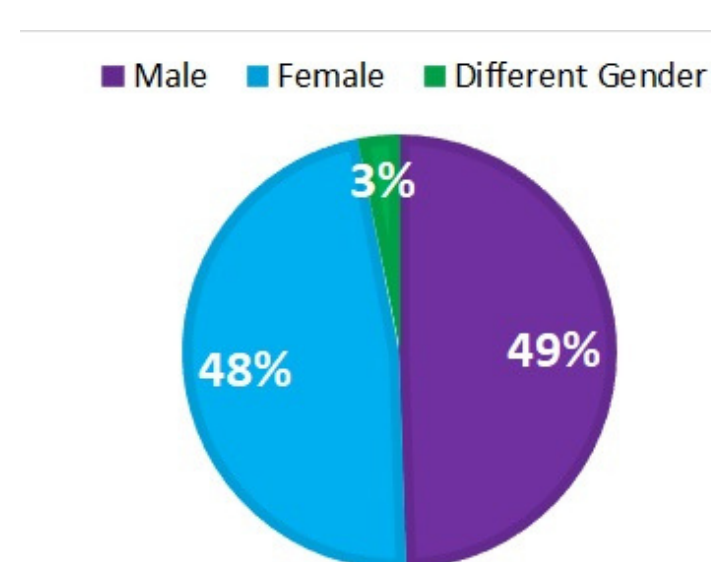
HYPOTHESIS

In order to establish the construct validity of the Manual BART Measure, we made the following predictions for content- and criterion-related validity

Content Validity	Criterion-Related Validity
Overall Riskiness	Vaping (Fatemi et al., 2021)
Impulsiveness	Sleep (Massar, 2021)
(Whiteside et al., 2001)	
Sensation Seeking	Spending Habits
(Hoyle et al., 2002)	(Wong & Carducci, 1991)
	Caffeine Consumption
	(Temple et al., 2017)
	Procrastination
	(Delibalta & Akbay, 2020)
	Tattoos (Carroll et al., 2002)
	Competitiveness
	(Cardenas et al., 2012)
	Dating

Additionally, based on previous research (Lejuez et al., 2002), we anticipated that males would demonstrate riskier behaviors and tendencies than females.

SAMPLE



N = 99
Average Age: 21.14
SD: 5.4
Range: 18–53

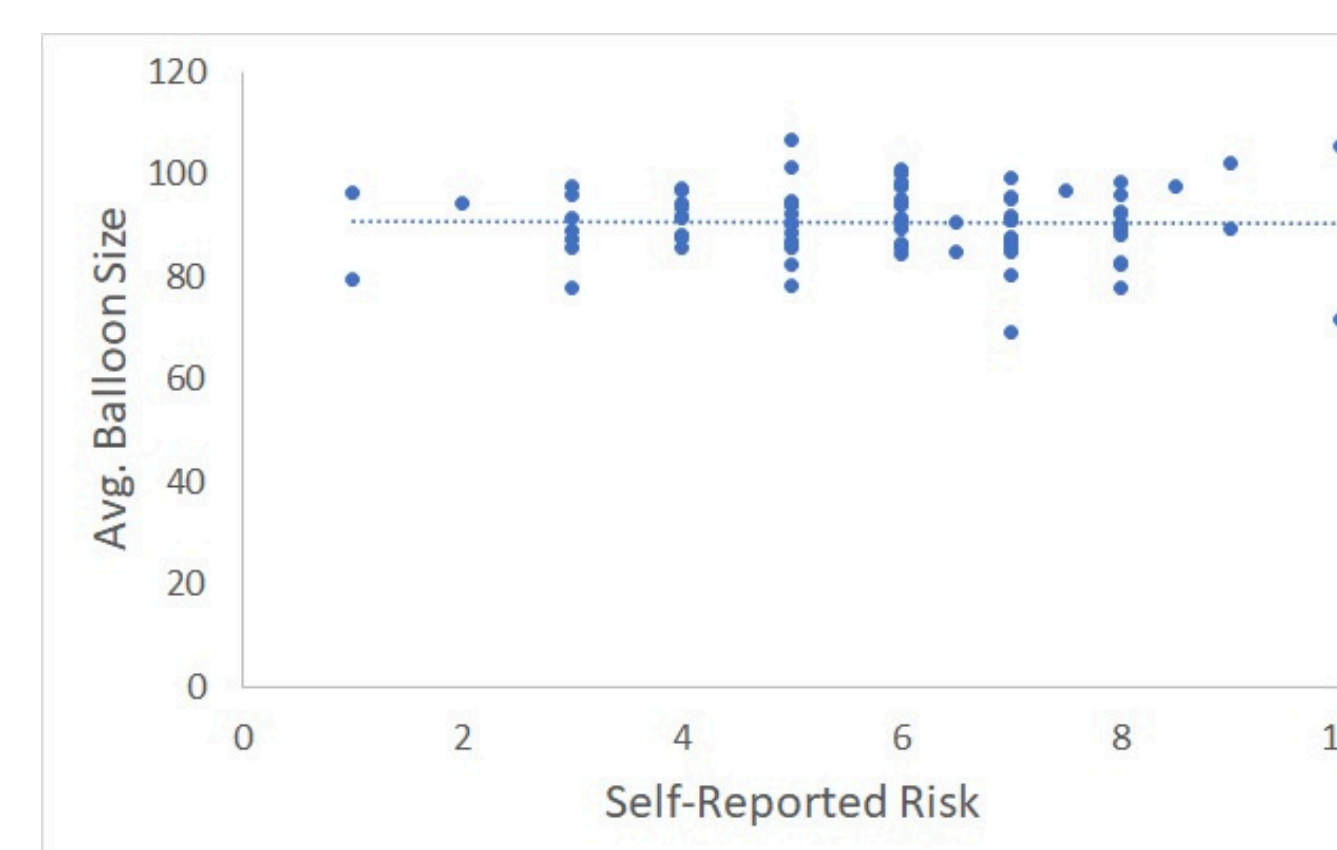
METHODS AND MEASURES



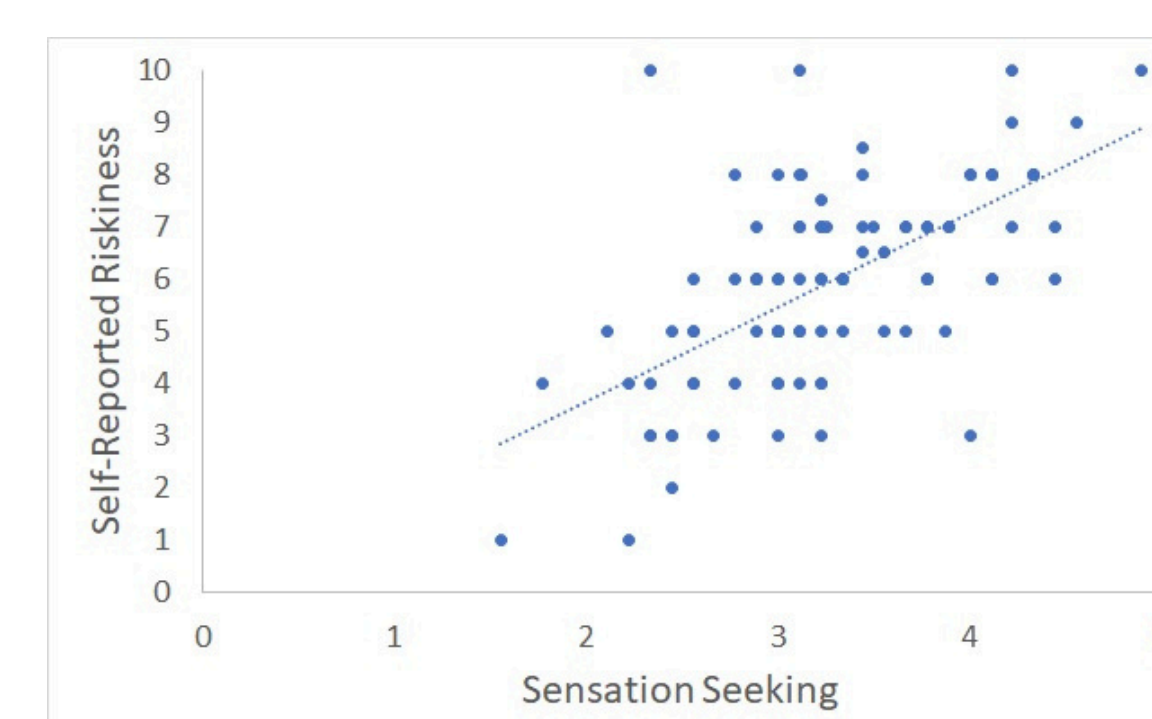
- Participants at GSU Atlanta campus took a survey that measured constructs of impulsiveness, sensation seeking, and self-reported risk-taking tendencies
- Other questions asked about caffeine consumption, vaping, etc. – behaviors thought to be associated with risk-taking.
- Using a deflated balloon and an air pump, students were instructed to inflate the balloon as large as they wanted.
- A \$20 gift card was offered for those who had the two largest balloons.
- If an individual popped their balloon, they were not eligible to win the gift cards.

Measure	Example
Balloon Size	Measured in cm. Inter-rater reliability, $r = .98$
Overall Riskiness	How risky would you describe yourself being (1 – 10)?
Impulsiveness	7 items ($\alpha = .740$); Ex.: I do things without thinking.
Sensation Seeking	9 items ($\alpha = .776$); Ex.: I would like to explore strange places.
Vaping	How often do you vape (never, less than monthly, monthly, weekly, daily, several times a day)?
Sleep	How many hours of sleep last night?
Spending Habits	I set aside money in advance for things I want to buy.
Caffeine Consumption	How many caffeinated beverages (coffee, tea, soda, or energy drinks) do you consume in a day?
Procrastination	I usually wait until the last minute to complete my school assignments.
Competitiveness	I am competitive person.
Tattoos	How many tattoos do you have?
Dating	I am likely to ask someone out on a date if I am interested in them.

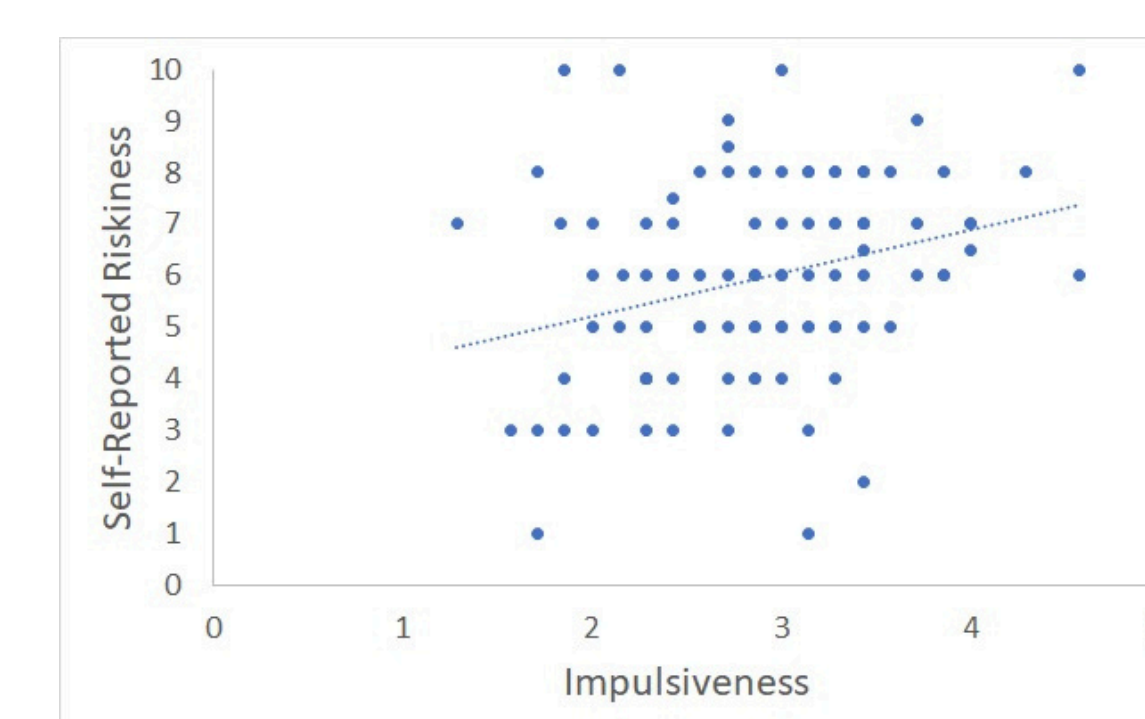
Convergent Validity



There was no relationship between balloon size and self-reported risk, $r(85) = -.01$, $p = .95$. Balloon size was also not correlated to Sensation Seeking [$r(86) = .20$, $p = .07$] nor Impulsiveness [$r(86) = .08$, $p = .48$]



Self-reported riskiness significantly correlated with Sensation Seeking [$r(96) = .62$, $p < .001$] as well as Impulsiveness [$r(96) = .29$, $p = .004$], which conceptually replicates previous work.



Criterion-Related Validity

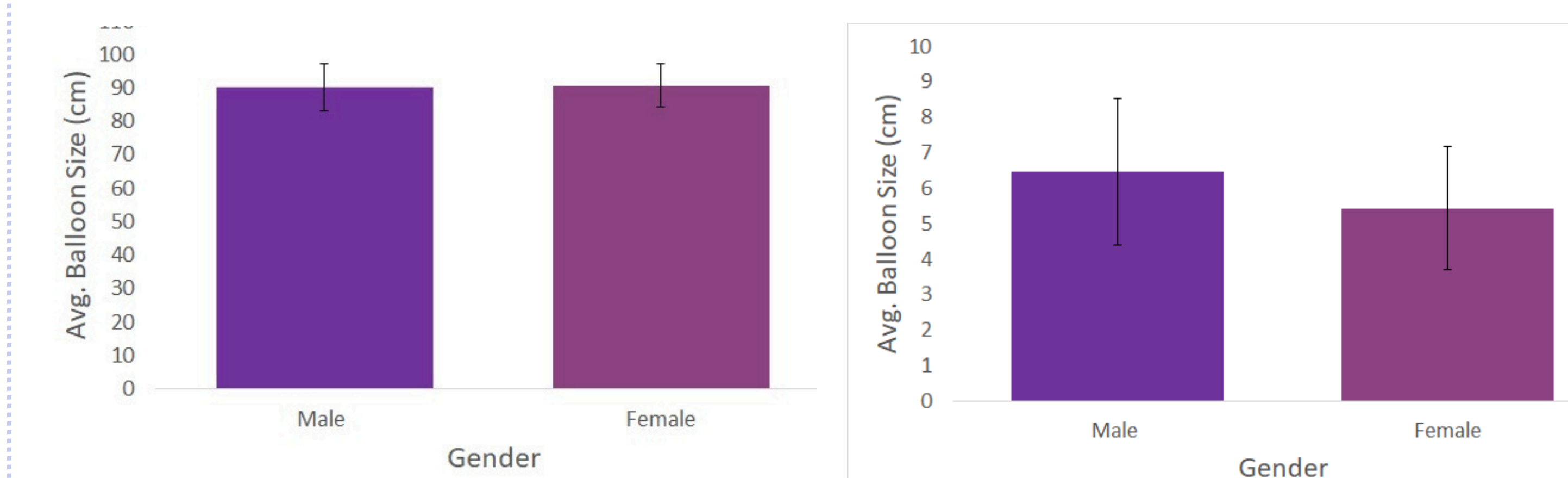
Measure	Balloon Size	Self-Reported Risk
Sleep	.24	-.22
Caffeine	.01	.08
Vaping	-.12	.12
Tattoos	.08	-.24 **
Competitiveness	.14	.09
Spending Habits	.08	.05
Asking on Dates	-.01	.26 **
Procrastination	.03	.01

We were not able to establish criterion-related validity with the manual BART measure. Behaviors that correlated with the computerized BART in previous research did not correlate with balloon size in our sample.

Self-reported riskiness significantly correlated with the number of tattoos, which is consistent with prior research. Sleep was weakly correlated with riskiness as well, but this fell just short of statistical significance.

Riskiness correlated with willingness to ask people on dates, which was also anticipated.

Gender Differences



Males and females inflated balloons to similar sizes, $t(83) = 0.43$, $p = .98$, $d = .1$. However, males self-reported being riskier than females, $t(92) = 2.65$, $p = .01$, $d = .55$, consistent with previous research

Balloons popped on 11.3% of attempts. We did not find any relationship between popped balloons and gender, $\chi^2(2) = 5.04$, $p = .081$, which was not in line with previous research.

CONCLUSIONS

Our results failed to establish the content- and criterion-related validity that we had predicted.

Associations with self-reported riskiness mostly conformed to expectations. It was associated with Impulsivity and Sensation Seeking, as well as the number of tattoos and willingness to ask people on dates.

However, our questions for other measures, such as vaping and caffeine consumption, for example, failed to show the correlations that we wanted.

Future Directions:

Use individual testing
Multiple opportunities to inflate balloons
Stricter control on observer effects